

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I VAISHNAVI.S (192521313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

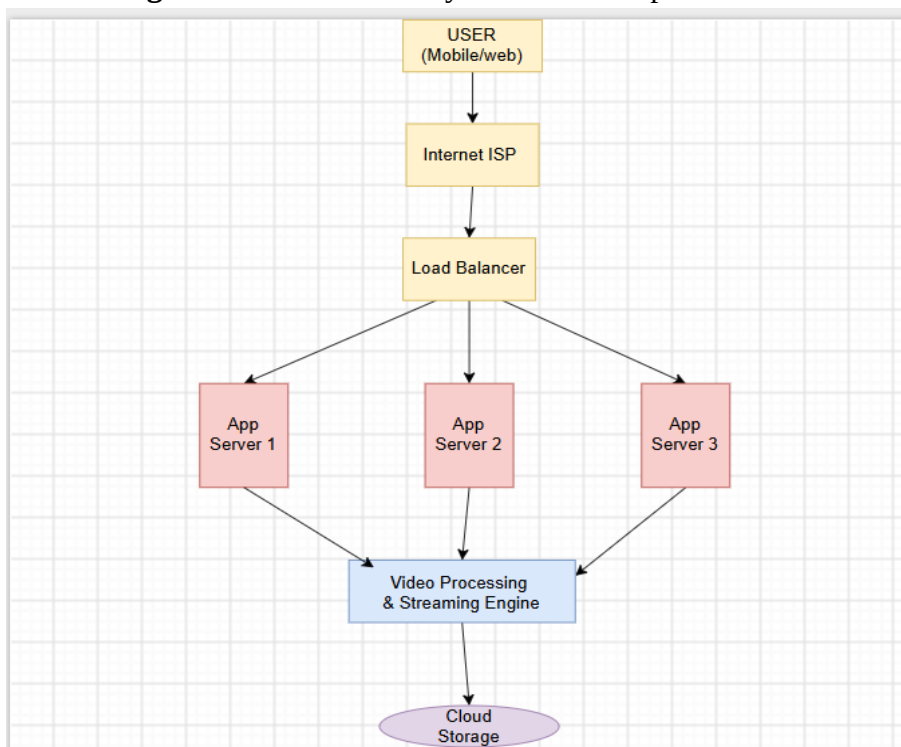
1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Cloud Architecture Design

The media company can use a cloud-based architecture to provide video streaming services to a large number of users. The architecture should be designed in such a way that it can easily handle heavy traffic without affecting the viewing experience.

The main components of the architecture are:

- **Users:** People watch videos using mobile phones, laptops, tablets, or smart TVs.
- **Load Balancer:** It receives all user requests and distributes them equally to different application servers.
- **Application Servers:** These servers process user requests such as login, video selection, and account management.
- **Auto Scaling Group:** It automatically increases or decreases the number of servers depending on the number of users.
- **Cache (CDN):** Frequently watched videos are stored in nearby cache servers so that users can access them quickly.
- **Cloud Storage:** Stores all video files safely with high durability.
- **Database:** Stores user details, subscriptions, watch history, and other information.
- **Monitoring Service:** Continuously checks server performance and automatically sends alerts.



Role of Load Balancing

Load balancing helps to distribute incoming requests among multiple application servers.

- Prevents one server from becoming overloaded.
- Improves system performance.
- Increases availability of the application.
- If one server fails, requests are redirected to another server.
- Ensures users experience smooth video streaming.

Role of Auto Scaling

Auto scaling automatically adjusts the number of servers according to user demand.

- During peak hours, additional servers are created automatically.
- During low traffic, extra servers are removed.
- Maintains fast response time.
- Prevents server overload.
- Reduces unnecessary cloud costs.

Cost Optimization Strategies

To reduce the overall cost while maintaining good performance, the company can use the following methods:

1. Use **Auto Scaling** so that servers run only when required.
2. Store popular videos in **CDN cache**, which reduces repeated access to the main storage.
3. Store rarely watched videos in lower-cost cloud storage.
4. Monitor cloud resource usage and remove unused resources.
5. Use reserved or long-term cloud instances for continuously running services to reduce pricing.
6. Compress video files to reduce storage space and bandwidth usage.

Conclusion

By using cloud storage, load balancing, auto scaling, and caching, the media company can provide smooth video streaming even for millions of users. These technologies improve performance, reduce buffering, increase availability, and also help in reducing cloud costs.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Solution Using Container Orchestration and Multi-Cloud:

A fintech startup can use container orchestration to manage applications running on different cloud providers. Containers package the application along with all the required files and dependencies, so the application behaves the same in every cloud environment. A container orchestration platform such as Kubernetes can be used to deploy, monitor, and manage containers automatically. It helps in scheduling containers, restarting failed containers, balancing workloads, and updating applications without stopping the service. In a multi-cloud strategy, the startup can deploy its application across two or more cloud providers. If one cloud experiences a problem, the application can continue running from another cloud, providing better availability and reliability.

Service Discovery

Service discovery helps containers find and communicate with each other automatically.

- Every service gets its own unique name.
- Containers can communicate without using fixed IP addresses.
- If a container is restarted or moved, communication continues without manual changes.
- This improves flexibility and reduces configuration work.

Scaling

Scaling allows the application to handle different numbers of users.

- During high traffic, Kubernetes creates additional container instances.
- During low traffic, extra containers are removed automatically.
- Load is shared among all running containers.
- This improves application performance and avoids resource wastage.

Benefits of Multi-Cloud Deployment:

- Improves application availability.
- Reduces dependency on a single cloud provider.
- Better disaster recovery.
- Users can connect to the nearest cloud, reducing response time.
- Different cloud providers can be selected based on cost and services.

Risks of Multi-Cloud Deployment

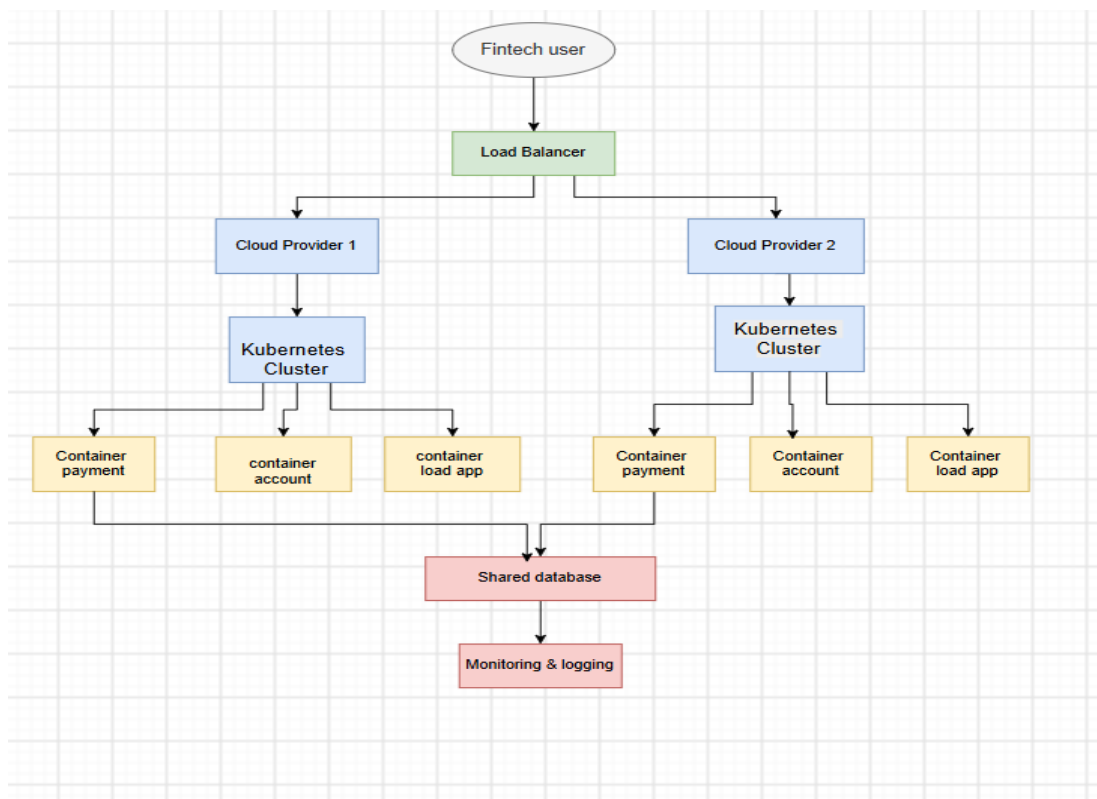
- Managing multiple cloud environments is more complex.
- Security policies must be maintained across all providers.
- Data synchronization between clouds can be difficult.
- Monitoring and troubleshooting require additional effort.
- Operational costs may increase if resources are not managed properly.

Need for Multi-Cloud Strategy

A fintech company handles financial transactions that require high availability. If one cloud provider becomes unavailable due to maintenance or failure, the application can continue running on another cloud provider. This helps avoid service interruptions and improves customer trust.

Conclusion

Using containers, Kubernetes, and a multi-cloud deployment model, the fintech startup can manage applications more efficiently. Container orchestration provides automated deployment, service discovery, and scaling, while multi-cloud improves availability and reliability. Proper monitoring and security practices are necessary to reduce the challenges of managing multiple cloud platforms.



3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Cloud Backup and Disaster Recovery Solution:

An e-learning platform stores important information such as student records, course materials, videos, assignments, and examination data. If data is accidentally deleted, the organization should have a proper backup and disaster recovery plan to restore the lost data quickly. The cloud solution should include automatic backup, data replication, and versioning. Automatic backups create copies of data at regular intervals. Replication stores the same data in another cloud region or data center. Versioning keeps multiple versions of the same file so that deleted or modified files can be recovered easily.

1. Backup

- Data is backed up automatically every day.
- Backup copies are stored safely in cloud storage.
- Lost files can be restored whenever needed.

2. Replication

- Data is copied to another cloud region.
- If the primary region fails, the secondary region continues the service.
- This improves data availability.

3. Versioning

- Multiple versions of each file are stored.
- If a file is deleted or edited by mistake, the previous version can be recovered.
- It protects important documents from accidental deletion.

Recovery Time Objective (RTO):

RTO is the maximum time required to restore the application after a failure. Example: If the RTO is 30 minutes, the e-learning platform should become operational within 30 minutes after the failure

- Automatic failover
- Fast backup restoration
- Replicated servers
- Disaster recovery site

Recovery Point Objective (RPO):

RPO is the maximum amount of data that can be lost during a failure. Example: If backups are taken every 15 minutes, only the last 15 minutes of data may be lost.

- Frequent backups
- Continuous data replication
- Database synchronization
- Cloud storage versioning

To Ensure Business Continuity:

1. Schedule automatic backups regularly.
2. Enable file versioning for all important data.
3. Replicate data to another cloud region.
4. Monitor cloud resources continuously.
5. Test disaster recovery plans periodically.
6. Use automatic failover during server failures.
7. Restore services from backup if data is lost.
8. Keep backup data encrypted for security.

Advantages:

- Prevents permanent data loss.
- Faster recovery after failures.
- High availability of learning services.
- Protects against accidental deletion.
- Improves reliability and business continuity.

Conclusion:

By using automatic backup, data replication, versioning, and disaster recovery, the e-learning platform can recover lost data quickly and continue providing services without major interruptions. These techniques reduce downtime, protect valuable information, and ensure continuous learning for students.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Hybrid Cloud Architecture:

A government agency stores both sensitive and non-sensitive information. Sensitive data, such as citizen records and confidential documents, should remain in the on-premises data center because of government regulations. Non-sensitive applications, such as public websites and online services, can be deployed in the public cloud. The on-premises data center and public cloud should be connected through a secure private connection or VPN. This allows data to move safely between the two environments. Identity management and encryption should also be used to protect users and data from unauthorized access.

Secure Communication:

The communication between the on-premises data center and the cloud should be protected using a VPN or private dedicated network.

- Secure transfer of information.
- Protection from unauthorized access.
- Safe communication between applications.
- Reliable connection for government services.

Identity Management:

- Using a single login for all applications.
- Providing role-based access to employees.
- Enabling multi-factor authentication (MFA).
- Regularly reviewing user permissions.
- Removing access for inactive users.

Two types of encryption should be used:

1. Data at Rest

- Encrypt files stored in databases and cloud storage.
- Protects stored government records.

2. Data in Transit

- Encrypt data while it is transferred between the on-premises data center and the cloud.
- Uses secure communication protocols.

Challenges in Hybrid Cloud Management:

1. Managing security across different environments.
2. Maintaining consistent user access policies.
3. Monitoring both cloud and on-premises resources.
4. Integrating different technologies.
5. Higher maintenance and operational costs.
6. Ensuring compliance with government regulations.
7. Data synchronization between environments.

Advantages of Hybrid Cloud:

- Sensitive data remains secure on-premises.
- Public services can scale using the cloud.
- Better flexibility and resource utilization.
- Improved disaster recovery.
- Cost-effective for non-sensitive applications.
- Meets government compliance requirements.

Conclusion:

A hybrid cloud model combines the security of on-premises infrastructure with the flexibility of the public cloud. Secure communication, identity management, and encryption protect government data, while proper monitoring and management help ensure reliable and compliant services.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Multi-Region Deployment Strategy

A SaaS (Software as a Service) provider serves users from different locations. To provide continuous service, the application should be deployed in multiple cloud regions. One region acts as the primary region, while another region acts as the secondary region. Both regions contain the same application and data. If the primary region becomes unavailable due to a network issue, hardware failure, or natural disaster, the traffic is automatically redirected to the secondary region. This ensures that users can continue using the application without major interruption.

Multi-Region Deployment

- Deploy the application in two or more cloud regions.
- Keep identical application servers in each region.
- Store copies of data in all regions.
- Use a global load balancer to direct users to the nearest available region.
- If one region fails, another region automatically handles user requests.

Data Replication

Data replication means creating copies of data in multiple regions. Replication can be done continuously so that the secondary region always contains the latest data.

- Keep data synchronized.
- Prevent data loss.
- Improve availability.
- Support quick recovery during failures.

Failover Mechanism

Failover automatically switches user traffic from the failed region to the healthy region.

1. Monitor the health of the primary region.
2. Detect server or network failure.
3. Redirect traffic to the secondary region.
4. Continue application services without interruption.
5. Restore the primary region after the issue is resolved.

Monitoring and Alerting

Monitoring continuously checks the health of cloud resources.

Important metrics include:

- CPU usage
- Memory usage
- Network traffic
- Disk utilization
- Application response time
- Server availability
- Database performance

Alerting sends notifications whenever a problem occurs.

Alerts can be sent through:

- Email
- SMS
- Mobile notifications
- Dashboard notifications

Early alerts help administrators fix problems before users are affected.

Benefits

- High availability of the application.
- Fault tolerance during regional failures.
- Reduced downtime.
- Better user experience.
- Improved disaster recovery.
- Reliable access for users from different locations.

Challenges

- Higher cloud infrastructure cost.
- Data synchronization between regions.
- More complex system management.
- Network latency during replication.
- Continuous monitoring is required.

Conclusion

A multi-region deployment improves the reliability and availability of SaaS applications. By using data replication, automatic failover, and continuous monitoring, the application can continue running even if one cloud region becomes unavailable. This ensures uninterrupted service and better customer satisfaction.